

Module 8. Deep Learning

- ❑ Motivation
- ❑ Improve Training of Deep Learning Models 
- ❑ Convolutional Neural Networks
- ❑ Recurrent Neural Networks
- ❑ Summary

Techniques to Improve Deep Learning Training

- ❑ Key Challenges for Training Deep Learning Models
- ❑ Responsive Activation Functions
- ❑ Adaptive Learning Rate
- ❑ Dropout
- ❑ Pretraining
- ❑ Cross Entropy

Key Challenges

❑ Optimization Problem in Deep Learning

- ❑ Minimize the (approximated) training error E
- ❑ (Stochastic) gradient descent to find the model parameter

$$E(\theta) = \frac{1}{m} \sum_{l=1}^m \text{Loss}(\hat{T}(X^l, \theta), T^l)$$

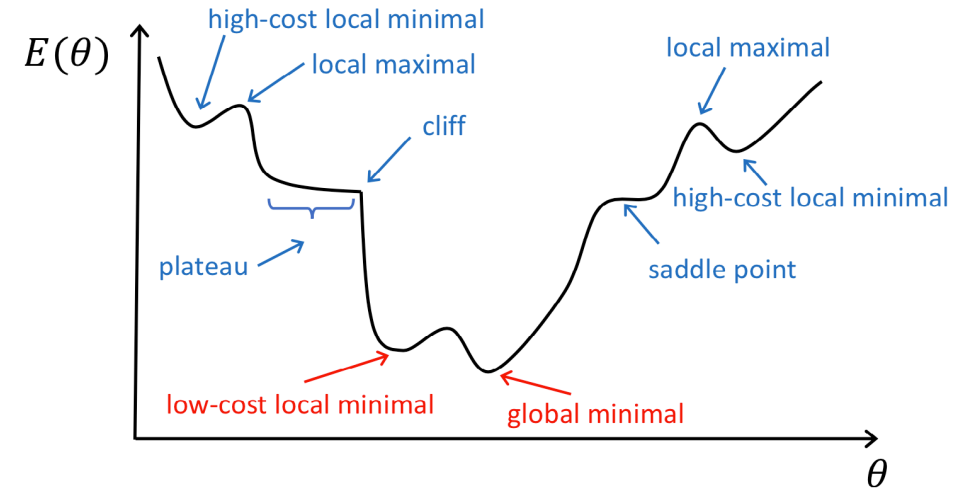
$$\theta_{t+1} = \theta_t - \eta g_t$$

❑ Challenge 1: Optimization

- ❑ E is non-convex in general
- ❑ How to find a high-quality local optima

❑ Challenge 2: Generalization

- ❑ What we do: minimize (approximated) training error
- ❑ What we really want: minimize the generalization error
- ❑ How to mitigate over-fitting



Responsive Activation Function

□ Saturation of Sigmoid Activation Function

- The output $\bar{O} = \sigma(I) = \frac{1}{1+e^{-I}} \in (0, 1)$
- The derivative $\frac{\partial O}{\partial I} = O(1 - O)$
- The error of an output unit $\delta_j = O_j(1 - O_j)(O_j - T_j)$

decaying

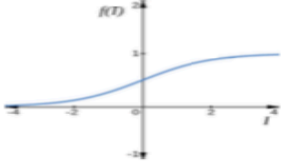
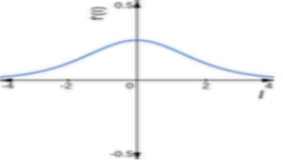
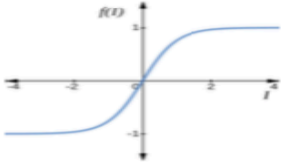
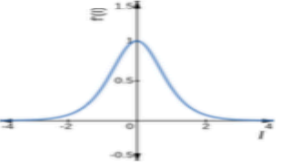
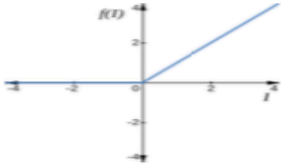
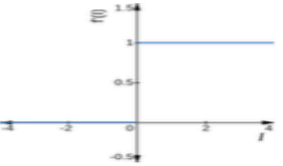
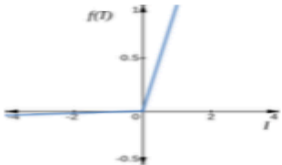
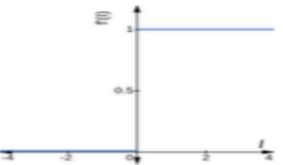
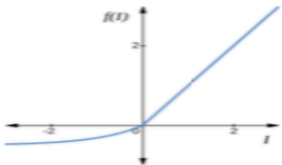
□ Saturation of Sigmoid Activation Function

- If $O_j \approx 1$ or $O_j \approx 0$, both derivative and error will be close to 0
- Further exacerbated due to backpropagation $\rightarrow$ gradient vanishing $\rightarrow$ B.P. is stuck or takes long time to terminate

□ A More Responsive Activation Function: Rectified Linear Unit (RELU)

- The output $O=f(I) = I$ if $I>0$, $O=0$ otherwise
- The gradient: $\frac{\partial O}{\partial I} = 1$ if $I > 0$ and $\frac{\partial O}{\partial I} = 0$
- The error: 0 if the unit is inactive ($I<0$), otherwise, aggregate all error terms from the units in the next higher layer the unit is connected to, w/o decaying $\rightarrow$ avoid gradient vanishing

Activation Functions

Name	Definition ($f(I)$)	Plot	Derivative of $f(I)$	Plot
Sigmoid	$\frac{1}{1+e^{-I}}$		$f(I)(1 - f(I))$	
Tanh	$\frac{e^I - e^{-I}}{e^I + e^{-I}}$		$1 - f(I)^2$	
ReLU	$\begin{cases} 0, & I \leq 0 \\ I, & I > 0 \end{cases}$		$\begin{cases} 0, & I \leq 0 \\ 1, & I \geq 0 \end{cases}$	
Leaky ReLU	$\begin{cases} 0.01 \times I, & I < 0 \\ I, & I \geq 0 \end{cases}$		$\begin{cases} 0.01, & I < 0 \\ 1, & I \geq 0 \end{cases}$	
ELU	$\begin{cases} \alpha(e^I - 1), & I \leq 0 \\ I, & I > 0 \end{cases}$		$\begin{cases} \alpha e^I, & I \leq 0 \\ 1, & I > 0 \end{cases}$	